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BIT 2114 Object Oriented Analysis and Design

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This presentation is intended to be covered within one week. The notes, examples and exercises should be supplemented with a good textbook. Most of the exercises have solutions/answers appearing elsewhere and accessible by clicking the green **Exercise** tag. To move back to the same page click the same tag appearing at the end of the solution/answer.

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LESSON 6

Requirements Analysis

Learning outcomes

Upon completing this topic, you should be able to:

1. Explain the different ways of classifying requirements.
2. Describe functional and non functional requirements.
3. Identify hard and soft constraints.



6.1. Requirements Analysis

Requirements analysis is the process of understanding the problem and the workable solution. It represents what is to be built and not how it is to be built and analyses users' wants and needs and communicates this in a language understood by the customer.

The unstructured collection of requirements are analysed and organized into coherent clusters such as functional and non-functional. Constraints and assumptions that would be considered in building the system are also considered. To understand the system context more, the domain is modelled at different levels of abstraction.

Use case models have been used in the previous chapter to model the system's behaviour from the user's point of view and



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to define the system at an abstract level.

Class diagrams can be used to capture the structure of the domain and sequence diagrams can be used to capture the dynamic behaviour of the domain.

Use-cases allow us to describe at a high level the basic functionality of a system in terms of the various actors who need to interact with the system. We now consider how to flesh out those use-cases, and to link them into the object modelling method we considered earlier.

From the use-cases, which describe broad functionality, we begin to analyse the details of the functionality of the systems we want to construct. Before doing the domain modelling, we have to classify our requirements that have been captured.



6.1.1. Requirements Classification

The requirements discovered from the previous sections are now classified into functional, non-functional and constraints. Assumptions which are made are also discussed herein.

6.1.2. Functional Requirements

Functional requirements describe the intended behavior of the system and specify tasks the system is to achieve.

Sommerville (2005) state that functional requirements describe the functionality or behavior of the system: services, reactions to input, exceptions and modes of operation. These functional requirements have been captured in the use case model.

1. A user shall log into the system using a unique ID and password.



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2. The system shall display an error alert if the log-in details are incorrect.
3. The user shall be able to log out.
4. The system shall be able to display the timetable for the whole week or for days or times specified by the user.
5. The system shall display the following information on the timetable.
 - (a) Time information.
 - (b) Days of the week.
 - (c) Course code.
 - (d) Classroom code.
6. The system shall be able to display the times for a course, for days or times specified by the user.



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7. The system shall be able to display the room assigned to a course, for days or times specified by the user.
8. The system shall be able to display the title and surname of instructors assigned to a course.
9. The system shall be able to display free rooms, for days or times specified by the user.
10. The system shall display a message if there are no free rooms for the days or times specified by the user.
11. The system shall be able to display free times, for days or periods specified by the user.
12. The system shall display a message if there are no free times for the days or period specified by the user.
13. The system shall be able to display courses offered by re-



spective departments.

14. The system shall be able to display courses respective student years.
15. The system shall be able to display the number of classes taking place at a time or day specified by the user.
16. The system shall provide the option for printing the queried timetable.
17. If the user opts to print the queried timetable, the system shall display an alert if no printer is connected to the computer.
18. The system shall have the option for saving to a disk, the queried timetable.
19. If the user opts to save the queried timetable to a disk,



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the system shall display an alert if the specified disk is not connected to the computer.

20. The system shall accept input data that is required for scheduling the timetable.
21. The system shall allow the user to update the new or changed data into the timetable.
22. The system shall display an alert if there is new data that has been input and no update has been made.
23. The system shall alert the user if there is data that has been changed and no update has been made.
24. The system shall display an alert for any data that is missing or conflicting, and is required to schedule the timetable.



6.1.3. Non- Functional Requirements.

Non- functional requirements are attributes that the system should have and constraints that are placed on the system.

The requirements are grouped into Quality requirements and Constraints.

Quality Requirements

The quality requirements used here are defined by ISO 9126. This is an international standard for software quality.

1. Functionality

Functionality is the essential purpose of a software system. The functionality of the class timetabling system is further defined under the following classes.

a) Suitability



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- The timetabling system shall be used to schedule the class timetabling system at JKUAT
- The timetable should be created before the beginning of the respective trimester.
- The timetabling system shall allow more than one instructor to be assigned to a course.

1. Accurateness

- The timetabling system shall schedule classes with no conflicts.

1. Interoperability

- The system should run on the JKUAT LAN where it will be accessed from any workstation and terminal.

1. Security



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- The different user groups shall have different security levels.
- Only the IT department will issue log-in details and can Change a user's security level.
- The log in details for the timetabling system shall be different from the log in details to the LAN.

b) Reliability

Reliability defines the capability of the system to maintain its service provision under defined conditions for certain periods of time.

1. Fault tolerance

- If a user enters their log-in details incorrectly, the timetabling system shall allow the user to re-enter their details as many



times until they get it right.

- The system shall automatically log out the user after 10 minutes of inactivity.
- The timetabling system shall check for typographical errors.

1. Recoverability

- If the LAN connection is lost, once regained, the timetabling system should be able to operate as normal.

1. Availability

- Unless the LAN is down, the system shall be online 24 hours a day, seven days a week.

c) Usability

1. Understability



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- The system shall contain a comprehensive user manual.
- The system shall display information in a simple form style.
- The system shall use the 12-hr clock system.

1. Learnability

- Users of all levels of expertise should not take more than a day to learn to use the system.

1. Operability

- The system shall have a well-defined and easy to use graphical user interface.
- The user shall interact with the system using a keyboard and a mouse.

d) Efficiency





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- The timetabling system shall respond to any user query in less than 10 seconds.
 - The timetabling system shall respond to user queries 99% of the time.
 - The system shall store data in a database system on the central server.
- e) Maintainability
- The system shall be modular; hence there shall be little effort in applying changes to the system.
- f) Portability
- The system shall be able to be used on PC and Macintosh platforms.
- g) Design



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- The system shall be developed using one or more platform-independent modular programming languages.

Constraints

Some constraints cannot be violated and have to be implemented into the system (hard constraints), while some can be violated but this violation should be avoided as much as possible (soft constraints). Therefore the constraints on JKUAT's class timetabling problem are divided into hard and soft.

Hard Constraints

1. The courses to be scheduled are undergraduate courses only.
2. Classes are scheduled on Monday to Friday only.



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3. Times for scheduling are from 8 am to 6 pm
4. No classes are to be scheduled during lunch hour between 1 pm and 2 pm.
5. The class meetings of two different courses with the same student enrolment cannot be assigned to the same time slot.
6. A lecturer cannot teach more than one class meeting at the same time slot.
7. The assigned number of class meetings that are scheduled for the same time slot cannot exceed the number of available classrooms.
8. All class meetings should be assigned to a time slot.
9. All class meeting should be assigned to a room.



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10. A room can only be assigned at most one class meeting at a time.
11. Course should be assigned to rooms with the necessary facilities.
12. Departmental constraints shall be implemented, for example, times for compulsory departmental meetings.
13. Time slot used is 1-hour for 2 credit hour courses, and 2 hours for 4 credit hour courses.
14. Courses should be assigned appropriate size rooms.
15. The Computer Laboratory 6 shall only be assigned practical courses of the Information Technology department.



Soft Constraints

These soft constraints are numbered in order of preference from high priority to lowest priority. During implementation, the scheduling algorithm may require that these constraints are weighted to determine which ones to consider first.

1. Where the number of students exceeds the appropriate room, more hours are allocated to the course.
2. Minimize consecutive classes for similar student groups or for the same instructor.
3. Departments should submit their preferences and constraints electronically.
4. There shall be minimal classes on Fridays afternoon.
5. There shall be minimal classes on Fridays in the afternoon



6. A lunch hour to be scheduled.

Assumptions.

Assumptions are conditions that must be true for use cases to terminate successfully. The assumptions made while developing the above requirements for JKUAT are:

1. The departments assign and register courses to students.
2. The departments identify who teaches the courses.
3. The courses for credit transfer students are scheduled at the same time as the courses for regular students. That is, while departments forward courses to be scheduled, the courses to be offered to credit transfer students should be included.
4. The maximum number of classes that a student can attend



in a day shall be specified.

5. The maximum number of hours that an instructor can teach in a day shall be specified.
6. The number of extra hours to be assigned to courses whose numbers exceed the capacity of the appropriate rooms, shall be specified.
7. User training shall be provided.
8. Unlike in the current system where individual departments forward complete timetables to the timetabling master for compilation, we shall assume that the departments shall only forward the following information to the timetabling master:

(a) Courses to be scheduled.



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- (b) Students registered for the courses.
- (c) Number of students in the courses.
- (d) Instructors assigned to the courses.
- (e) Special room requirements such as computer lab, science lab etc.
- (f) Special times requirements such as availability of part time lecturers.
- (g) Departmental hard constraints, For example, no classes to be scheduled on Tuesday 2.00 pm to 4.00 pm for the computer science department since they have departmental meetings at that time.



6.2. Summary

Classifying requirements into the above groups assists in extracting external qualities of the system and the properties that these qualities shall have. Constraints define what requirements the system must fulfil for hard constraints and which ones it can fulfill if possible, for soft constraints. Assumptions are made to accommodate information that is needed for the requirements to be fulfilled.



Learning Activities

Identify all functional, non-functional requirements for the Travel Agent System.

Revision Questions

Example . What general advantages are claimed for the use of a methodology in the development of information systems?

Solution: for revision



EXERCISE 1.  What does it mean, to say that most modern methodologies are agile?

EXERCISE 2.  How might the following contribute to reuse in systems development? a) Encapsulation. b) Generalisation. c) Composition. d) Patterns.



References and Additional Reading Materials

1. CADALYST: A popular magazine focusing on AutoCAD.
<http://www.cadonline.com>
2. An electronic CADD magazine (e-zine) focusing on the CADD industry news. You can subscribe this weekly e-zine free of charge. <http://www.upfrontezine.com/>